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Falling Interest Rates and Credit Reallocation: Lessons from General Equilibrium

Contents

1 Big picture

- **Question.** What does regulator-induced bank consolidation do to credit supply, loan pricing, screening, and ex post loan performance?
- **Setting.** The paper studies the restructuring of the Spanish savings-banks sector between 2009 and 2011, when the number of savings banks fell from 37 to 12 after the crisis.
- **Core comparison.** The key empirical contrast is between consolidations executed as full mergers and acquisitions (M&As) and consolidations executed as *sistemas institucionales de protección* (SIPs), a business-group structure.
- **Why this comparison is useful.** M&A banks become centralized entities that can coordinate lending decisions and exercise market power more strongly. SIP banks share liquidity and solvency support and some centralized functions, but remain legally separate entities, report separately to the regulator, and are less able to coordinate lending policies.
- **Reduced-form result.** Relative to SIP banks, merged banks cut credit supply, charge higher interest-rate spreads, reject fewer loan applicants, and report lower nonperforming-loan ratios.
- **Main puzzle.** Merged banks appear to lend *less*, yet they also reject *fewer* applicants and end up with *better* ex post performance. So quantity, standards, and performance move in different directions.
- **Structural answer.** The paper estimates a model in which banks jointly choose loan rates and lending standards. Consolidation increases market power, but it also sharply lowers screening costs. As a result, merged banks can lend to somewhat riskier borrowers while still reducing realized bad loans.
- **Bottom line.** Credit performance improves after merger consolidation not because merged banks become tighter on every margin, but because higher rates and better screening dominate the looser extensive-margin approval policy.
- **Why the paper matters.** It links industrial organization, banking, and crisis policy. It shows that consolidation can shrink credit quantity while improving private bank profitability and ex post loan performance, with important implications for public restructuring programs.

2 Data and measurement (key objects)

2.1 Institutional setting: the Spanish consolidation program

- After the 2008 crisis, Spain passed Royal Decree 9/2009 to restructure the savings-banks sector.
- Savings banks could obtain public capital from the Fondo de Reestructuración Ordenada Bancaria (FROB) in exchange for submitting a consolidation plan.
- The law allowed two forms of consolidation:
  - **M&A.** A standard merger that creates a single centralized bank.
  - **SIP.** A business-group structure with mutual liquidity and solvency support, but the constituent banks remain separate legal entities.

- SIPs resemble mergers in capital regulation, risk-sharing, and access to pooled borrower information, but differ in one critical way: they preserve legal independence and separate reporting to the credit registry.
- That legal independence matters because it weakens the coordination of lending policies within the group and limits the scope for exercising market power.

2.2 Main data

- **Banco de España Central Credit Registry (CCR).** Monthly bank-firm loan exposures above 6,000 euros, including drawn and undrawn credit lines.
- **NPL information.** The CCR reports the amount of nonperforming loans for each bank-firm pair.
- **Loan applications proxy.** The CCR records banks' requests for information about firms' credit situations. Following the literature, the paper interprets such requests as loan applications by new borrowers.
- **Interest rates.** Supervisory data provide average interest rates on new loans at the bank-month level, not at the borrower level.
- **Bank balance sheets.** Banco de España supervisory data are used to construct bank size, leverage, profitability, and FROB-capital controls.
- **Firm balance sheets.** The Integrated Central Balance Sheet Data Office Survey (CBSDO) provides total assets, leverage, liquidity, profitability, and industry information for nonfinancial corporations.

2.3 Sample construction

- Main reduced-form sample: November 2007 to November 2011.
- The paper studies the 37 savings banks that participated in the 12 consolidation operations between November 2009 and December 2010.
- These banks account for about 40% of total credit in Spain and around 90% of the credit extended by the savings-banks sector.
- The post period is truncated in November 2011 to avoid contamination from the later sovereign-debt-rescue phase.
- For M&As, postconsolidation data are observed at the group level because the preexisting entities disappear.
- For SIPs, banks continue reporting separately, so the paper aggregates the constituent banks into the group before and after consolidation when required.

2.4 Key variables

- **Credit quantity.** Total bank-firm credit, including credit lines; in the main regressions the dependent variable is the semiannual growth rate of log credit.

- **Rejected applications.** For new firm-bank pairs, an application is counted as rejected if a bank requests information but no increase in the bank-firm credit balance is observed in the current month or within the next 3 months.
- **Interest-rate spread.** The spread between the average nominal interest rate on newly issued loans and the 3-month Euribor.
- **Credit performance.** The ratio of nonperforming loans to total loans.
- **Instrument.** A historical measure of within-region credit overlap in 1995, based on the market share of the second-largest savings bank inside the future consolidation group.

2.5 Descriptive patterns that motivate the analysis

- The average bank group is large: mean assets are 83.13 billion euros, with a mean capital ratio of 5.60% and mean ROA of 0.42%.
- The average firm in the matched sample is small: mean assets are 1.90 million euros.
- From the pre- to the postconsolidation period, average semiannual credit growth falls from -0.05 to -0.13, the average NPL ratio rises from 1.80% to 3.09%, and average spreads rise from 1.84% to 2.73%.
- At the unconditional level, M&A and SIP banks look similar before the reform, then diverge afterwards: merged banks lend less, charge more, and report lower NPL ratios.

3 Models and empirical specifications (all core equations + interpretation)

3.1 Instrument construction: historical credit overlap

The historical overlap measure is

$$Credit\ Overlap_{jm} = \frac{Total\ Credit_{b'm}}{\sum_{b\in m} Total\ Credit_{bm}}, \tag{1}$$

where b' is the second-largest savings bank in future group j in province m.

- **Interpretation.** This measures how much the future members of a consolidation group overlapped in a province back in 1995, when savings banks were still concentrated in their historical areas of influence.
- **Why it matters.** Higher overlap in the home region is interpreted as stronger political pressure for a within-region merger rather than a SIP structure.

3.2 Baseline reduced-form specification for credit

The main quasi-experimental regression is

$$\begin{aligned} y_{jit} = {} & \alpha_1 M_j + \alpha_2 Post_{jt} + \alpha_3 (M_j \times Post_{jt}) \\ & + \beta X_{j,t-h} + \gamma Z_{i,t-h} + \zeta FROB_{jt} + \delta_{kmst} + \varepsilon_{jit}, \end{aligned} \tag{2}$$

where y_{jit} is the semiannual growth rate of log credit, M_j is an M&A-group indicator, $Post_{jt}$ switches on after consolidation, $X_{j,t-h}$ are lagged bank controls, $Z_{i,t-h}$ are lagged firm controls, $FROB_{jt}$ controls for public capital injections, and δ_{kmst} are industry-location-size-time fixed effects.

- **Coefficient of interest.** α_3 is the differential postconsolidation effect for merged banks relative to SIP banks.
- **Interpretation.** This design compares the same types of firms across bank groups, while controlling for common shocks at the industry-province-size-time level.

3.3 Interest-rate specification

For loan pricing, the paper estimates

$$w_{jt} = \alpha_1 M_j + \alpha_2 Post_{jt} + \alpha_3 (M_j \times Post_{jt}) + \beta X_{j,t-h} + \zeta FROB_{jt} + \iota_{jt}, \tag{3}$$

where w_{jt} is the spread between the average nominal interest rate on new loans and the 3-month Euribor.

- **Interpretation.** This is the pricing counterpart to Equation (2), estimated at the bank-month level because loan-rate data are only available at that aggregation.

3.4 NPL specification

To study ex post loan performance for postconsolidation new relationships, the paper estimates

$$z_{jit} = \alpha M_j + \beta X_{j,t-h} + \gamma Z_{i,t-h} + \zeta FROB_{jt} + \delta_{kmst} + \varepsilon_{jit}, \tag{4}$$

where z_{jit} is the proportion of NPLs over total loans.

- **Interpretation.** Because the sample is restricted to bank-firm pairs with no preexisting relationship at the start of consolidation, the dependent variable cleanly reflects loans originated in the post period.

3.5 Borrower demand in the structural model

Borrower i's indirect utility from borrowing from bank b in province m and month t is

$$U^D_{ibmt} = \alpha^D P_{bt} + \beta^D X_{bt} + \tau^D_b + \omega^D_{mt} + \xi^D_{bmt} + \varepsilon^D_{ibmt}. \tag{5}$$

- P_{bt} is the bank-month loan rate and X_{bt} includes bank size.
- τ^D_b are bank fixed effects, ω^D_{mt} are province-month fixed effects, and ξ^D_{bmt} is an unobserved bank-province-month demand shifter.
- **Interpretation.** Banks are differentiated products in a standard empirical-IO sense; borrowers choose among them and an outside option made of fringe banks.

3.6 Screening and acceptance probability

Bank b's acceptance probability for borrower i is

$$\phi_{ibt} = \frac{\exp(\gamma_{bt} + \lambda_{ibt})}{1 + \exp(\gamma_{bt} + \lambda_{ibt})}. \tag{6}$$

- γ_{bt} is the bank-month lending-standards parameter.
- λ_{ibt} is a borrower-bank-month component.
- **Interpretation.** A higher γ_{bt} means looser lending standards and a higher probability that an application is accepted.

Aggregating over possible borrower choice sets gives bank b's market share:

$$s_{bmt} = \sum_{S \in C_{bt}} \prod_{l \in S} \phi_{lt} \prod_{k \notin S} (1 - \phi_{kt}) \frac{\exp(\delta^D_{bmt})}{1 + \sum_{r \in S} \exp(\delta^D_{rmt})}. \tag{7}$$

- **Interpretation.** Market shares depend jointly on pricing and screening: P_{bt} affects demand directly through δ^D_{bmt}, while γ_{bt} affects how likely the bank is to belong to the borrower's feasible set.

3.7 Expected default risk

Borrower i's indirect utility from defaulting is

$$U^F_{ibt} = \alpha^F P_{bt} + \beta^F X_{bt} + \tau^F_b + \omega^F_t + \xi^F_{bt} + \varepsilon^F_{ibt}. \tag{8}$$

This implies the bank-month default share

$$F_{bt} = \frac{\exp(\alpha^F P_{bt} + \beta^F X_{bt} + \tau^F_b + \omega^F_t + \xi^F_{bt})}{1 + \exp(\alpha^F P_{bt} + \beta^F X_{bt} + \tau^F_b + \omega^F_t + \xi^F_{bt})}. \tag{9}$$

- **Interpretation.** Higher rates can attract borrowers with higher expected default risk, conditional on the lending standards the bank sets.

3.8 Bank profits and cost structure

Banks choose rates and lending standards to maximize expected profits:

$$\Pi_{bt} = \left[(1 + P_{bt})(1 - F_{bt}) + R_{bt} F_{bt} - C_{bt}\right] Q_{bt}, \tag{10}$$

where $Q_{bt} = \sum_m s_{bmt} M_{mt}$ is credit quantity and R_{bt} is the recovery rate on defaulted loans. Average costs are

$$C_{bt} = \exp(c_{0bt} + c_{1bt} \gamma_{bt}). \tag{11}$$

- c_{0bt} is the baseline operating-cost component.
- c_{1bt} is the screening-cost component.
- **Interpretation.** If $c_{1bt} > 0$, relaxing lending standards increases the marginal cost of lending because the bank extends credit to riskier borrowers. A fall in c_{1bt} means screening becomes cheaper or better.

3.9 First-order conditions and recovered cost parameters

The first-order condition with respect to interest rates is

$$\begin{aligned}\frac{\partial \Pi_{bt}}{\partial P_{bt}} &= [(1+P_{bt})(1-F_{bt}) + R_{bt}F_{bt} - C_{bt}]\frac{\partial Q_{bt}}{\partial P_{bt}} \\ &\quad + \Big[1 - F_{bt} + (R_{bt} - 1 - P_{bt})\frac{\partial F_{bt}}{\partial P_{bt}}\Big]Q_{bt} = 0.\end{aligned} \tag{12}$$

The first-order condition with respect to lending standards is

$$\frac{\partial \Pi_{bt}}{\partial \gamma_{bt}} = [(1+P_{bt})(1-F_{bt}) + R_{bt}F_{bt} - C_{bt}]\frac{\partial Q_{bt}}{\partial \gamma_{bt}} - \frac{\partial C_{bt}}{\partial \gamma_{bt}}Q_{bt} = 0. \tag{13}$$

These imply recovered average costs

$$\widehat{C}_{bt} = (1+P_{bt})(1-F_{bt}) + R_{bt}F_{bt} + \Big[1 - F_{bt} + (R_{bt} - 1 - P_{bt})\frac{\partial F_{bt}}{\partial P_{bt}}\Big]\frac{Q_{bt}}{\partial Q_{bt}/\partial P_{bt}}, \tag{14}$$

and recovered cost slope

$$\frac{\partial \widehat{C}_{bt}}{\partial \widehat{\gamma}_{bt}} = \Big[(1+P_{bt})(1-F_{bt}) + R_{bt}F_{bt} - \widehat{C}_{bt}\Big]\frac{\partial Q_{bt}/\partial \gamma_{bt}}{Q_{bt}}. \tag{15}$$

Then the cost parameters are recovered as

$$\widehat{c}_{1bt} = \frac{\partial \widehat{C}_{bt}/\partial \widehat{\gamma}_{bt}}{\widehat{C}_{bt}}, \tag{16}$$

$$\widehat{c}_{0bt} = \log(\widehat{C}_{bt}) - \widehat{c}_{1bt}\widehat{\gamma}_{bt}. \tag{17}$$

- **Interpretation.** These equations let the authors back out both the overall cost level and the extra cost of loosening standards from equilibrium behavior.
- **Key empirical point.** The data imply that $\widehat{c}_{1bt}$ is positive in the baseline, but shifts downward strongly after merger consolidation.

3.10 Estimating the screening shock

The paper first estimates lending-standard shocks using loan-application data:

$$\phi_{ibt} = \gamma_{bt} + \rho Y_{it} + \zeta_{kst} + \eta_{mst} + \epsilon_{ibt}, \tag{18}$$

where Y_{it} are borrower financial variables, ζ_{kst} are sector-size-time fixed effects, and η_{mst} are location-size-time fixed effects.

Then it constructs

$$\widehat{\lambda}_{bt} = \log\Big(\frac{\phi_{bt}}{1-\phi_{bt}}\Big) - \widehat{\gamma}_{bt}. \tag{19}$$

- **Interpretation.** The application data let the authors separate borrower demand for credit from bank-side screening decisions, which is the central innovation needed to model acceptance rates.

3.11 Simulated borrower choice probabilities

For a simulated borrower ι, the probability of choosing bank b conditional on the feasible set S^{ι} is

$$P^{\iota}_{bmt} = \frac{\exp(\delta^{D}_{bmt})}{1 + \sum_{r\in S^{\iota}} \exp(\delta^{D}_{rmt})}. \tag{20}$$

- **Interpretation.** The paper simulates borrower choice sets because banks choose whether to admit borrowers in the first place.

3.12 Default estimation equation

Expected default is estimated through

$$\log\Big(\frac{F_{bt}}{1-F_{bt}}\Big) = \alpha^{F} P_{bt} + \beta^{F} X_{bt} + \tau^{F}_{b} + \omega^{F}_{t} + \xi^{F}_{bt}. \tag{21}$$

- **Identification.** Interest rates in the demand and default equations are instrumented with lagged NPLs; the paper also reports similar demand elasticity when using household deposits as an alternative cost-shock instrument.

4 Identification and instruments (what makes the estimates credible)

4.1 Why M&A versus SIP identifies market-power effects

- Both M&As and SIPs are forms of regulator-induced consolidation and therefore share many restructuring features.
- The paper shows that M&A and SIP banks have similar operating-cost dynamics before and after consolidation.
- The key difference is organizational: M&As centralize decision-making, while SIPs remain legally separate and are still exposed to antitrust constraints if they collude.
- Therefore, comparing M&As to SIPs is meant to isolate the effect of stronger internal coordination and market power rather than the generic effect of consolidation.

4.2 Evidence that SIP groups are less coordinated

- M&A banks use hard information more intensively: per thousand euros of credit granted, they access the credit registry about 15% more often than SIP banks.
- Within a given SIP, individual banks often treat the same firm's application differently.
- Table 2 shows that, in four out of six SIP groups, bank-specific effects are statistically significant in a regression with SIP-group-firm fixed effects.
- This is direct evidence that lending policies are not fully coordinated inside SIP groups.

4.3 Instrument for the form of consolidation

- The endogenous choice is whether a consolidation is executed as an M&A or as a SIP.
- The instrument is a binary indicator for whether historical credit overlap in the group's main region is above the median.
- The economic logic is political: regional governments preferred within-region mergers to keep headquarters local, and 1995 overlap captures proximity to the historical regional sphere of influence.
- Using 1995 overlap is attractive because it predates the crisis and the consolidation program by many years.

4.4 Threats and how the paper handles them

- **Concern:** M&A banks may differ from SIP banks on unobservables related to future lending behavior.
- **Response:** The paper controls for bank balance-sheet variables, firm balance-sheet variables, FROB capital injections, and high-dimensional industry-location-size-time fixed effects.
- **Concern:** Changes in firm demand could drive the results.
- **Response:** Industry-location-size-time fixed effects are designed to absorb local sectoral demand shocks faced by comparable firms.
- **Concern:** The NPL regression may mix old and new loans.
- **Response:** The NPL analysis is restricted to firm-bank pairs with no relationship at the start of the consolidation period.

5 Tables and figures

5.1 Table 1: Summary statistics (quasi-experimental analysis)

Read:

- The bank groups are economically large, but the borrower sample is dominated by small firms.
- Comparing panels C and D, credit growth becomes more negative after consolidation, interest-rate spreads rise, and NPL ratios increase at the aggregate level, reflecting the harsh macro environment.
- The decline in rejected applications in the raw data is important: it foreshadows that merged banks may be tightening on price while loosening on approval standards.

5.2 Figure 1: Unconditional evidence

Read:

- Before the law in November 2009, M&A and SIP groups look broadly similar in credit, spreads, and NPLs.

Table 1
Summary statistics: Quasi-experimental analysis

A. Bank groups

Variables	November 2007–November 2011					
	Mean	Median	Standard deviation	5th Percentile	95th Percentile	N
TA (BE)	83.13	47.73	83.40	19.73	280.46	60
Capital ratio (%)	5.60	4.94	1.71	3.67	8.88	60
ROA (%)	0.42	0.45	0.28	0.05	0.79	60

B. Firms

Variables	November 2007–November 2011					
	Mean	Median	Standard deviation	5th Percentile	95th Percentile	N
TA (ME)	1.90	0.47	5.42	0.04	6.95	1,035,318
Total liabilities/TA (%)	72.74	79.90	26.29	19.14	100.00	1,035,318
Liquid assets/TA (%)	9.34	3.32	14.86	0.00	41.59	1,035,318
ROA (%)	3.86	5.13	19.05	−24.54	27.74	1,035,318

C. Bank-firm relationships

Variables	November 2007–Consolidation date					
	Mean	Median	Standard deviation	5th Percentile	95th Percentile	N
Δ *log(Credit)*	−0.05	−0.02	1.38	−2.40	2.77	1,666,166
NPL (%)	1.80	0.00	13.03	0.00	0.00	1,666,166
Rejected applications (Count)	0.42	0	0.59	0	1	174,696
Interest rate spread (%)	1.84	1.61	0.88	0.67	3.39	354

D. Bank-firm relationships

Variables	Consolidation date–November 2011					
	Mean	Median	Standard deviation	5th Percentile	95th Percentile	N
Δ *log(Credit)*	−0.13	−0.04	1.19	−2.26	1.06	936,838
NPL (%)	3.09	0.00	17.02	0.00	0.00	936,838
Rejected applications (Count)	0.27	0	0.51	0	1	68,449
Interest rate spread (%)	2.73	2.78	0.58	1.71	3.56	234

This table contains descriptive statistics (mean, median, standard deviation, 5th and 95th percentiles, and number of observations [N]) for bank and firm characteristics (panels A and B, respectively) as well as for firm-bank credit balances (panels C and D). Panel A reports end-of-year information at the level of each individual group j, panel B at the level of individual firms-year, and panels C and D are at the bank-firm-semester level for credit change, number of rejected applications, and NPLs, and at the bank-month level for interest rate spreads. Panels A and B report the statistics between November 2007 and November 2011. Panel C reports descriptive statistics on the semiannual change in the credit balance between November 2007 and the date in which consolidation took place, on the number of rejected applications received by a firm from the savings banks within a given group in each semester, on the ratio of NPLs over total loans for the whole sample of firm-bank pairs at the end of each semester, and the average monthly interest rate spread over the 3-month Euribor at the bank-month level. Panel D does the same for the period between the date of consolidation and November 2011. Firm-level variables and the log change in credit are winsorized such that we set the observations above (below) the 99% (1%) percentiles at the value of the 99% (1%) percentile. For additional information on the construction of these variables, see the Internet Appendix.

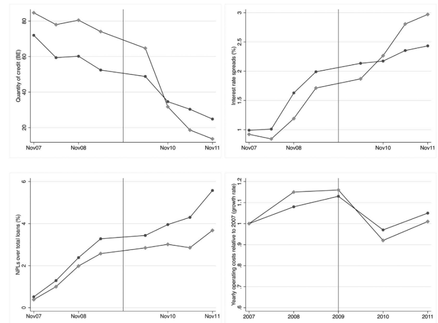

Figure 1
Unconditional evidence
In the top-left, top-right, and bottom-left panels, we report the semiannual pattern of the quantity of credit, the interest rate spreads relative to the 3-month Euribor, and the NPLs over total loans for M&A banks (grey line, squares) and SIP banks (black line, circles) in the period ranging between four semesters before and four semesters after the month in which the Law was passed (November 2009). In the bottom-right panel, we report the yearly operating costs relative to 2007 for M&A banks (grey line, squares) and SIP banks (black line, circles) in the 2 years before and the 2 years after 2009. Since the information is aggregated at the group level, we do not plot the confidence intervals. For information on the construction of the variables, see the Internet Appendix.

- After the reform, M&A groups lend less and charge higher spreads than SIP groups.
- The NPL ratio of M&A groups rises less than that of SIP groups, while operating-cost paths remain very similar. This visual combination motivates the paper's claim that market power and screening efficiency, not simple operating-cost cuts, are central.

5.3 Table 2: Uncoordinated lending policies across SIP banks

Read:

- Within several SIP groups, individual savings banks treat the same borrower's application differently even after group formation.
- Statistically significant coefficients for SIP groups 2, 3, 4, and 6 reject the idea of perfectly centralized screening within SIPs.
- This is a key institutional validation for using SIP groups as a lower-market-power benchmark.

5.4 Table 3: Credit overlap and consolidation form

Read:

- The coefficient on historical credit overlap is positive and highly significant in all specifications: 3.823, 3.535, and 4.289.

Table 2
Uncoordinated lending policies across SIP banks

	Rejected application
Savings bank in SIP group 1	0.333
	[0.320]
Savings bank in SIP group 2	0.432***
	[0.114]
Savings bank in SIP group 3	0.467***
	[0.111]
Savings bank in SIP group 4	0.324***
	[0.062]
Savings bank in SIP group 5	0.338
	[0.270]
Savings bank in SIP group 6	0.141**
	[0.069]
Observations	1,518
R-squared	.677
SIP Group-Firm FE	YES

In this table, we test whether savings banks within a given SIP have similar lending policies after forming the group. We restrict our sample to the savings banks in our sample that formed an SIP. The analysis was conducted between the month of the formation of SIP group j and November 2011. Our dependent variable is a dummy variable equal to 1 if (1) a savings bank belonging to a given SIP requested information on a firm at any month after the group's formation and (2) we do not observe an increase in the firm-bank credit balance during this period. When these conditions are jointly satisfied, we infer that the firm loan application was rejected. We regress this variable on dummy variables for each specific savings bank (the omitted term refers to one of the savings banks in each given SIP) and on SIP group-firm fixed effects. We report the coefficient only for the savings bank dummy with the lowest p-value. We report robust standard errors in brackets. $^{*} p < .1$; $^{**} p < .05$; $^{***} p < .01$.

Table 3
Credit overlap and consolidation form

Variables	(1) All regions	(2) All regions	(3) "Main" regions
Credit overlap in 1995	3.823***	3.535***	4.289***
	[1.278]	[0.983]	[1.305]
Observations	330	330	55
R-squared	.464	.432	.364
Province FE	YES	NO	NO
Bank Controls	YES	YES	YES

The table reports the results of the regressions that relate the dummy variable that is equal to 1 for M&As and 0 for SIPs and a measure of credit overlap in 1995 at the province level as defined in Equation (1). In columns 1 and 2, we use all the regions with provinces in which the savings banks were active in 1995, whereas in column 3 we use province-level data from the "main" region of each group. In all the columns, we use the following set of bank controls: *log(TA)*, *Capital ratio*, *ROA*, and *(FROB funds)/TA*. Moreover, in column 1, we use province fixed effects. Robust standard errors (in brackets) are clustered at the province level. $^{*} p < .1$; $^{**} p < .05$; $^{***} p < .01$.

- The result holds with and without province fixed effects and when restricting attention to the group's "main" region.
- This is the paper's first-stage evidence that regional political geography helps explain whether consolidation took the form of an M&A or a SIP.

5.5 Table 4: The effect of market power on credit supply and performance

Read:

- **Credit.** The interaction $Post \times M\&A$ is -0.032, implying about a 3% larger semiannual credit contraction for merged banks, roughly 16,000 euros less credit per firm per semester.
- **Standards.** The same interaction is -0.111 in the rejected-applications regression, meaning M&A banks reject fewer applications, about 25% of the average count.
- **Pricing.** The interest-rate interaction is 0.758, which the paper maps into roughly 31 basis points higher spreads, about 9% of the average baseline spread.
- **Performance.** In the postconsolidation NPL regression, the M&A coefficient is -0.011, about 1.1 percentage points lower NPLs, or around 35% of the post-period mean.
- **Core message.** Mergers reduce quantity, raise prices, loosen acceptance margins, and yet improve ex post loan performance.

5.6 Table 5: Structural-analysis summary statistics

Read:

- The average market share at the bank-province-month level is 2.49%, with mean loan rates of 4.59% and mean expected default risk of 7.95%.
- The firm sample used to estimate supply remains skewed toward smaller firms, and the mean acceptance rate for applications is 59.06%.
- These data are rich enough to jointly discipline demand, screening, and default behavior.

Table 4
The effect of market power on credit supply and performance

	(1) Δ *log(Credit)*	(2) *# Rejected applications*	(3) *Interest rates*	(4) *NPLs*
Post	0.001	–0.031	0.265	
	[0.025]	[0.046]	[0.234]	
M&A	–0.010	–0.055*	–0.447	–0.011***
	[0.021]	[0.033]	[0.361]	[0.004]
Post × *M&A*	–0.032*	–0.111**	0.758*	
	[0.019]	[0.056]	[0.351]	
Observations	2,603,004	243,145	588	121,003
Industry-Location-Size-Time FE	YES	YES	–	YES
Bank Controls	YES	YES	YES	YES
Firm Controls	YES	YES	–	YES

This table reports the results obtained from three instrumental variable regression analyses in which the instrument is a binary variable that indicates whether the credit overlap in the bank group region of influence is high. The dependent variable in column 1 is the semiannual growth rate of the (log) volume of total credit between November 2007 and November 2011. Column 2 shows the results obtained from a regression analysis analogous to that in column 1 but using as the dependent variable the number of rejected applications received by a firm from the savings banks within a given group j in each semester. Column 3 reports the results obtained from a regression analysis in which the dependent variable is the spread of the average monthly interest rate charged by a given bank group j to new loans granted in month t to nonfinancial institutions over the 3-month Euribor. The information on interest rates is available for different categories of loan maturity (less than 1 year, between 1 and 5 years, and more than 5 years). We perform a regression analysis in which the interest rate is the weighted average across the three maturity buckets, using the new operations within each maturity bucket as weights so that the unit of observation is bank-month. In columns 1 to 3, the sample period goes from November 2007 to November 2011. In column 4, we report the results obtained in a regression analysis in which the dependent variable is the average proportion of NPLs at the end of every semester in the period spanning between the announcement of group j-formation (M&A or SIP) and November 2011. The explanatory variable of interest in columns 1 to 3 is the interaction of a dummy variable that is equal 1 if the savings bank chooses an M&A (and 0 if they choose an SIP) and a dummy variable that equals 1 starting from the end of the semester in which group j (M&A or SIP) formed. Since column 4 considers the bank-firm pairs with no credit relationship at the date of consolidation, the variable of interest is the dummy M&A, which is estimated only for the postconsolidation period. The set of control variables in all columns includes bank characteristics such as *log(TA)*, *Capital ratio*, *ROA*, and *(FROB funds)/TA*. Additionally, in columns 1, 2 and 4 we use industry-location-size-time fixed-effects and the following firm characteristics as control variables: *Total liabilities/TA*, *Liquidity/TA*, *ROA*, and *log(TA)*. Robust standard errors (in brackets) are clustered at industry-province-size and bank group-province level in columns 1, 2 and 4 and at the bank group and month level in column 3. * $p<.1$; ** $p<.05$; *** $p<.01$. For information on the construction of the variables, see the Internet Appendix.

Table 5
Summary statistics: Structural analysis

A. Descriptive statistics of the variables used for the estimation of demand

Variables	November 2007–November 2011					
	Mean	Median	Standard deviation	5th Percentile	95th Percentile	N
Market share (%)	2.49	0.39	4.62	<0.01	11.69	84,964
Interest rate (%)	4.59	4.52	1.28	2.45	6.61	3,119
Default risk (%)	7.95	6.96	4.09	4.03	16.34	3,119
TA (BE)	42.28	12.59	82.66	1.82	210.61	3,119

B. Firm and firm-bank variables used for the estimation of supply

Variables	November 2007–November 2011					
	Mean	Median	Standard deviation	5th Percentile	95th Percentile	N
TA (ME)	2.98	0.45	12.88	0.02	9.14	975,595
Total liabilities/TA (%)	68.06	75.66	29.06	8.64	100.00	975,595
Liquid assets/TA (%)	12.77	4.55	20.19	0.00	57.87	975,595
ROA (%)	4.42	5.55	21.24	−27.61	32.21	975,595
Accepted applications (%)	59.06	100.00	49.17	0.00	100.00	2,012,159

This table contains descriptive statistics (mean, median, standard deviation, 5th and 95th percentiles, and the number of observations) for bank characteristics (panel A) as well as for firm and firm-bank characteristics (panel B). In panel A, the variable *Market share* is aggregated at the bank-province-month level, and the variables *Interest rate* and *TA* are aggregated at the bank-month level. In panel B, we restrict the sample to firms that apply for new credit to banks with which they had no previous relationships. Firm characteristics (*TA*, *Total liabilities/TA*, *Liquid assets/TA*, and *ROA*) are aggregated at the firm-year level. In contrast, the share of accepted applications is aggregated at the firm-bank-month level. Panels A and B report the statistics between November 2007 and November 2011. For information on the construction of the variables, see the Internet Appendix.

5.7 Table 6: Demand and default estimation results

Read:

- The demand-side interest-rate coefficient is −0.53, implying a demand elasticity of about −2.37.
- The default equation has a positive interest-rate coefficient of 1.02, meaning higher rates are associated with riskier borrower pools.
- Bank size raises demand but does not meaningfully affect default risk in the default regression.

Table 6
Demand and default estimation results

Variables	Demand	Default
Interest rate	–0.53***	1.02**
	(0.11)	(0.48)
log(TA)	1.16***	0.16
	(0.06)	(0.10)
Observations	84,964	3,119
Bank FE	Yes	Yes
Province-Month FE	Yes	Yes

An observation is a bank-month-province for the demand model and a bank-month for the default model. The demand model corresponds to Equation (7); the default model, to Equation (9). Robust standard errors are in parentheses for the demand model; standard errors are clustered at the month level for the default model. * $p<.1$; ** $p<.05$; *** $p<.01$. For information on the construction of the variables, see the Internet Appendix.

Table 7
Descriptives of cost parameters

Variables	Mean	Standard deviation	10th Percentile	Median	90th Percentile	N
$\hat{c}_{0bt}$	0.96	0.75	0.18	1.13	1.63	3,119
$\hat{c}_{1bt}$	0.29	0.79	0.03	0.11	0.42	3,119

This table contains descriptive statistics (mean, median, standard deviation, 5th and 95th percentiles, and the number of observations) for parameter c_{0bt}, which proxies banks' operating costs, and parameter c_{1bt}, which proxies banks' screening costs. See Equation (11). An observation is a bank-month.

Table 8
Divestiture versus M&A without and with cost efficiencies, lending standards

	Divested v. No cost efficiencies	M&A banks Cost efficiencies	M&A banks No lending standards
%Δ *Credit quantity*	6.41	3.22	−1.61
%Δ *Interest rate*	0.30	−7.53	2.05
Δ *Acceptance rate*	0.1	−9.64	−0.21
Δ *Default risk*	−0.11	−2.69	0.59
%Δ *Profit*	−6.61	−22.97	−1.13

The table reports the percentage changes (for *Credit quantity*, *Interest rate*, and *Profit*) and percentage point changes (for *Acceptance rate* and *Default risk*) in the equilibrium outcomes for the divested M&A savings banks relative to the same savings banks once they form the M&A groups (in the first two columns), which instead make coordinated decisions on interest rates and acceptance rates, assuming that M&A consolidation either does not produce any cost efficiency (in the first column) or that it does (in the second column). In the third column, we compare M&A banks optimizing only over interest rates to M&A banks optimizing over both interest rates and lending standards. To calculate the equilibrium outcome variables for the divested banks, since we do not observe their market shares in the postconsolidation period, we compute a weighted average across the banks that form each M&A group, using as weights the market share of each savings bank within the group as predicted by the model. We treat all the savings banks within each group j formed via a M&A as a single entity. An observation is at the bank group-month level. These results refer only to the months after the first consolidation (November 2009).

5.8 Table 7: Descriptives of cost parameters

Read:

- The mean recovered operating-cost component is $\widehat{c}_{0bt} = 0.96$.
- The mean screening-cost slope is $\widehat{c}_{1bt} = 0.29$, and the paper stresses that $\widehat{c}_{1bt}$ is positive throughout the sample.
- Positive $\widehat{c}_{1bt}$ means that relaxing standards is normally costly because it rotates the average-cost curve inward.

5.9 Figure 2: Cost parameter c1bt with and without consolidation

Read:

- The distribution of the screening-cost parameter for merged banks with efficiencies is shifted sharply to the left relative to the counterfactual no-efficiency distribution.
- The Kolmogorov-Smirnov statistic is 0.4306 with $p = 0.000$, strongly rejecting equality of distributions.
- This figure is the clearest evidence that mergers materially reduce screening costs.

5.10 Table 8: Divestiture versus M&A, with and without efficiencies

Read:

- **Without cost efficiencies.** Divested banks lend 6.41% more and have 6.61% lower profits than the corresponding merged banks. The market-power effect alone lowers quantity and raises profits, but the interest-rate and acceptance-rate movements are modest.
- **With cost efficiencies.** Relative to merged banks, divested banks have 3.22% more credit, 7.53% lower rates, 9.64 percentage points lower acceptance, 2.69 percentage points lower default risk, and 22.97% lower profits. So the full merger effect is: less credit, higher rates, looser standards, more ex ante risk, and much higher profits.
- **No-lending-standards counterfactual.** If merged banks could set only rates, credit would be 1.61% lower, rates 2.05% higher, default risk 0.59 percentage points higher, and profits 1.13% lower. Modeling acceptance decisions is therefore quantitatively important.

5.11 Table 9: Credit risk borne by M&A banks

Read:

- The reduction in NPLs for merged banks is stronger for riskier firms.
- The M&A coefficient is -0.005 for firms below the median probability of default, -0.016 for firms above the median, -0.031 for firms with PD $> 3\%$, and -0.057 for firms with PD $> 5\%$.
- This is strong validation of the model's interpretation: merged banks take on more ex ante risk but screen riskier firms more effectively, so ex post bad loans still fall.

Table 9
Credit risk borne by M&A banks

Variables	(1) PD≤Median	(2) PD>Median	(3) PD>3%	(4) PD>5%
M&A	–0.005*	–0.016***	–0.031**	–0.057***
	[0.003]	[0.006]	[0.013]	[0.022]
Observations	45,115	44,925	14,059	4,656
Industry-Location-Size-Time FE	YES	YES	YES	YES
Bank Controls	YES	YES	YES	YES
Firm Controls	YES	YES	YES	YES

This table reports the results obtained from four instrumental variable regression analyses in which the instrument is a binary variable that indicates whether the credit overlap in the bank group region of influence is high. The dependent variable in columns 1-4 is the average proportion of NPLs at the end of every semester in the period spanning between the announcement of group j-formation (M&A or SIP) and November 2011. This regression analysis is conducted for various firms' subsamples categorized by risk levels. Columns 1 and 2 display results for firms with a probability of default at the time of loan issuance either below or above the median credit risk distribution for the sample. In columns 3 and 4, we narrow the sample to two groups of even riskier firms. Column 3 includes firms with a probability of default above 3%, corresponding to a credit quality step (CQS) category defined by the ECB of 7. In contrast, column 4 focuses on firms in the highest (riskiest) CQS, which is 8. The explanatory variable of interest is the dummy *M&A*, which is estimated only for the postconsolidation period since we consider the bank-firm pairs with no credit relationship as of the beginning of the consolidation period (i.e., November 2009). The set of control variables in all columns includes bank characteristics such as *log(TA)*, *Capital ratio*, *ROA*, and *FROB/TA*. Additionally, we use industry-location-size-time fixed effects and the following firm characteristics as control variables: *Total liabilities/TA*, *Liquidity/TA*, *ROA*, and *log(TA)*. Robust standard errors (in brackets) are clustered at industry-province-size and bank group-province level. * $p < .1$; ** $p < .05$; *** $p < .01$. For additional information, see the Internet Appendix.

6 Short summary

- **Conclusion.** Relative to SIP business groups, merger consolidation reduces credit quantity, raises loan rates, lowers rejection rates, and lowers realized NPLs.
- **Mechanism.** Mergers increase market power and, crucially, reduce screening costs, which lets banks loosen approval standards while still improving ex post performance.
- **Big lesson.** In banking, tighter credit quantity and better loan performance do not necessarily mean tighter standards. Prices, screening, and portfolio composition must be studied jointly.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- Regulator-induced bank consolidation has two separable effects: it changes competitive conduct and it changes internal efficiency.
- The reduced-form evidence shows that merged banks lend less, charge more, reject fewer applicants, and report fewer NPLs than SIP banks.
- The structural model explains these facts by combining a market-power channel with a screening-efficiency channel.
- Merged banks use higher rates to contract quantity and improve margins.
- At the same time, lower screening costs let them accept more borrowers, including riskier ones, without worsening ex post portfolio performance.
- The profits of merged banks rise materially; in the full counterfactual, divested banks would earn about 23% lower profits than merged banks.

7.2 Economic intuition

- A merger does not simply make banks "tougher." It changes the whole menu of supply decisions.

- When the bank can both price more aggressively and screen more efficiently, it may lend to a riskier pool at the extensive margin while still generating better realized outcomes.
- The paper's key insight is therefore about the distinction between *ex ante borrower risk* and *ex post loan performance*: mergers raise the former but improve the latter because screening gets better.

7.3 Takeaway

This paper is powerful because it combines a convincing institutional comparison with a structural model that matches the bank's actual decision problem. The empirical punchline is not just that consolidation reduces credit supply. It is that consolidation simultaneously shifts prices, approval policies, and screening technology. That is exactly why a merger can produce less lending, more private profitability, and fewer bad loans at the same time.